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Combining Mixed Reality and 2D Displays for 3D Design: An Exploratory Study to Enhance User Needs

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Abstract

This work explores how integrating Mixed Reality with conventional 2D displays affects 3D modeling workflows by addressing issues like limited depth perception and spatial awareness. A MR prototype was developed, leveraging a Meta Quest 3¹ headset, Unity's Mesh Sync for real-time updates, hand-based interactions, animation playback in 3D, and a virtual monitor to complement traditional interfaces. The prototype aimed to improve workflow efficiency and depth understanding without disrupting familiar modeling environments. User testing involved six experienced 3D modelers. Their feedback was evaluated through a combination of quantitative metrics, including task completion times and End User Computing Satisfaction scores, as well as qualitative insights from semi-structured interviews. The findings revealed that while task completion times increased when using the MR-enhanced setup, spatial accuracy and error detection were improved. The MR system reduced the need for frequent model orbiting and allowed participants to identify and address alignment issues more effectively.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; Empirical studies in HCI; Graphics input devices.

Keywords

Mixed Reality, 3D Design, Authoring Tools

ACM Reference Format:

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1 Introduction

The entertainment industry, particularly video games and movies, is heavily based on 3D graphics. In education and professional training, 3D graphics have become crucial for simulations and visualizations in fields such as medicine [6], engineering [27], and

the military [17], enhancing learning and training outcomes. In industries like automotive and aerospace, 3D modeling has become integral to the design and manufacturing process [16]. It allows for more accurate prototypes, better visualization of products, and more efficient production workflows, ultimately driving innovation and efficiency in these sectors.

Technologies such as Augmented Reality (AR) and Virtual Reality (VR) are paving the way for new applications in gaming, education, training, and more. Devices like the Apple Vision Pro² aim to represent the next step in making immersive 3D experiences more accessible. Advances in artificial intelligence and machine learning are further enhancing 3D graphics, enabling more realistic rendering, procedural generation of content, and automated design processes [23].

Although there are automated processes for creating 3D models and environments (such as photogrammetry and gaussian splatting), manual 3D modeling in tools like Maya or Blender is still an indispensable technique in the industry. The integration of Mixed Reality (MR) into traditional 3D modeling workflows builds upon these developments, offering new ways to visualize, interact with, and create complex digital models that bridge virtual and real environments.

Common problems in traditional 3D modeling workflows are widespread and include depth perception, user interface issues and performance constraints. **Depth Perception:** It can be difficult to accurately perceive depth and spatial relationships between objects. This can lead to mistakes in positioning and aligning components within the 3D space. **User Interface Issues:** The software interfaces for 3D modeling are often complex, and navigating through various tools, menus, and panels can be overwhelming. This can slow down the workflow and increase the learning curve. **Performance Constraints:** High-poly models and detailed scenes can strain the performance of the system, leading to lag and slower response times. This can hinder productivity and cause frustration.

The primary objective of this study is to investigate the impact of integrating MR into traditional 3D modeling workflows. Specifically, the research aims to determine whether the use of an MR headset, in conjunction with standard 2D monitors, affects user satisfaction and workflow efficiency for experienced 3D modelers. By developing a prototype with a combined 2D + MR setup, the study seeks to explore how MR influences key aspects of the modeling process, including spatial understanding, task accuracy, and design decision-making. Through both qualitative insights from expert interviews and quantitative metrics such as task performance

¹<https://www.meta.com/de/quest/quest-3/>



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²<https://www.apple.com/de/apple-vision-pro/>

and user satisfaction scores, the research aspires to provide an exploratory evaluation of the practical benefits and limitations of incorporating MR into professional 3D design environments.

2 Related Work

2.1 Depth perception

Depth perception is a complex visual process that allows individuals to perceive the distance and three-dimensional structure of objects in their environment. It relies on a combination of monocular and binocular cues, which are processed by the visual system to create a coherent perception of depth. Monocular depth cues are visual indicators that can be perceived with one eye. These cues include linear perspective, occlusion (interposition), size, texture gradients, shading, and motion parallax. For instance, linear perspective involves the convergence of parallel lines as they recede into the distance, while occlusion occurs when one object overlaps another, indicating that the overlapped object is farther away [22]. Motion parallax, which refers to the relative motion of objects as an observer moves, also provides significant depth information; closer objects appear to move faster than those further away [5, 15]. These monocular cues are essential for depth perception, especially in two-dimensional images where binocular cues are absent [12, 22]. In contrast, binocular depth perception relies on the slightly different views of the world that each eye receives, known as binocular disparity. This disparity arises because the eyes are spaced apart, allowing them to capture images from slightly different angles. The brain processes these differences to gauge depth, enhancing the perception of three-dimensionality [7, 8]. Research indicates that depth perception is generally more accurate under binocular conditions than monocular ones, as the combination of monocular and binocular cues can lead to a more reliable estimation of depth [7–9]. For example, studies have shown that binocular viewing improves the reliability and consistency of depth perception compared to monocular viewing alone [7, 15]. The interaction between monocular and binocular cues is crucial for effective depth perception. In many scenarios, these cues work together to enhance the overall perception of depth. For instance, when binocular disparity is distorted, monocular cues can help correct the perception of depth, demonstrating a compensatory effect [1, 19]. This interplay is particularly evident in complex visual environments, such as virtual reality, where both types of cues are present and can influence depth judgments [19, 20]. However, conflicts between monocular and binocular cues can lead to perceptual distortions, as seen in situations where the cues provide contradictory information about an object's depth [3, 21]. In summary, human depth perception is a multifaceted process that integrates monocular and binocular cues to create a coherent understanding of spatial relationships in the environment.

2.2 3D graphics design

Ivan Sutherland's Sketchpad from 1963 was one of the first CAD systems [2] and marked a major milestone in computer graphics, introducing the concept of interactive drawing. This system allowed users to manipulate graphical objects on a screen, effectively replacing traditional paper-based drafting methods [24, 25]. The transition from 2D drafting to 3D CAD systems revolutionized

the way complex designs were visualized and manufactured. Early 3D CAD systems like CATIA (Dassault Systèmes) and CADAM (Lockheed) were developed to meet the growing demands of industries requiring surface and solid modeling, especially in automotive and aerospace [10]. The entertainment industry began embracing 3D graphics for movies, video games, and animation. Tools like 3DS Max (Autodesk) and Maya became essential for 3D animation, visual effects, and game development.

The integration of 3D CAD with VR, AR, and MR is shaping the future of digital design. MR systems, such as Microsoft HoloLens³ and Apple Vision Pro⁴, are enabling professionals to interact with 3D models in real-world environments. Although there is no established workflow for 3D graphics design in MR yet, there are several approaches and tools that lead into this direction. For instance, the VR Scene Inspection add-on in Blender allows users to explore their 3D models in virtual reality for enhanced depth perception. Although primarily intended for viewing rather than full interaction, this tool provides a more immersive way to inspect 3D scenes. Enscape is a powerful real-time visualization and VR plugin that integrates seamlessly with Rhino 3D and other CAD software, enabling users to immerse themselves in their models. Furthermore, there are several standalone VR applications for 3D modelling such as Google Tilt Brush, Gravity Sketch or ShapesXR which do not yet offer the same scope of functions as established professional 2D-screen applications like Blender.

However, without widespread adoption of professional AR/VR solutions for modeling, professionals working with 3D models still rely on a limited set of strategies to compensate for the 2D screen's constraints. These methods often introduce their own drawbacks and inefficiencies:

Constant Orbiting Around Models: One of the most commonly used methods to maintain awareness of a model's three-dimensionality is to continuously orbit or rotate around the model after each modification. This technique helps in catching errors related to depth or perspective but is inherently time-consuming and can become distracting, pulling focus away from the creative or technical task. Moreover, this method is reactive, as it allows users to spot errors only after changes have been made, rather than preventing them in the first place.

Using Multiple Viewports: Another approach involves opening multiple viewing windows within the software, each displaying the model from different angles (typically orthographic views such as front, side, and top). This can help users understand spatial relationships more effectively but comes with its own set of limitations. The more views that are opened, the less screen space is available for each individual window, leading to crowded interfaces and smaller workspaces. Additionally, the process of constantly shifting focus between different views can interrupt workflow and reduce the intuitive flow of working on a single canvas.

2.3 2D vs. 3D

The conflict arising from using 2D devices, while being a 3D perceiving being has been discussed in the literature from many different

³<https://learn.microsoft.com/de-de/hololens/>

⁴<https://www.apple.com/de/apple-vision-pro/>

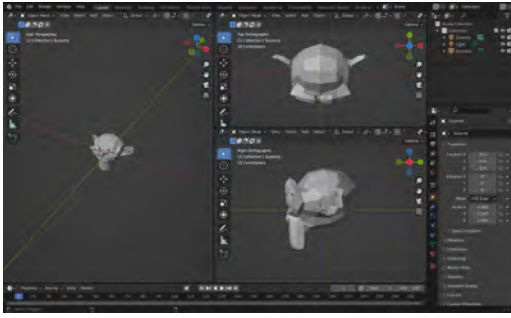


Figure 1: The 3D authoring tool Blender offers multiple view-ports to compensate for limited depth perception.

angles. The significant benefits of 3D content, compared to 2D content, in intrinsic motivation and learning was proven on a total of 565 student visiting Biology, Chemistry and Geology classes at a Czech middle- and high school [26]. Other research indicates mixed results when comparing the usage of 2D to 3D systems. Dima et al. reported mixed results in a Quality of Experience (QoE) study on stereoscopic 3D teleconferencing systems. While participants subjectively favored the QoE of the 2D system, objective measures indicated significantly better performance in depth-based tasks with the 3D system [4].

One of the main premises to overcome the 2D/ 3D perception problem is the ability to display 3D content to the user. VR, AR, MR headsets offer these capabilities and their ability to allow for stereoscopy has been proven in a study by [18]. They found out that VR and AR headsets induce similar errors in depth estimation [18]. Lin et al. went even further and discusses that egocentric distance estimation in virtual environments is about 80% accurate, compared to 94% in the real world. This reduction in accuracy can negatively affect depth perception and positioning tasks, potentially influencing user performance in virtual environments [13]. Since they released their investigation of this phenomenon in 2015, HMDs have evolved. Therefore, in 2022, Kelly et al. conducted a meta analysis based on 137 data points from 61 publications with 20 modern headsets [11]. They found out that depth perception in both VR and AR is consistently underestimated. This depth compression varies depending on several factors, including the type of task, the distance involved, and the technology used. Newer stereoscopic displays, higher frame rates, and better rendering techniques help mitigate, but not eliminate, depth perception errors [11]. On the topic of depth perception using stereoscopic headsets, a paper was published, discussing the vergence-accommodation conflict (AVC) in optical see-through displays. The authors explain the AVC and highlight how it still remains an obstacle in the development of high quality optical performance headsets [28].

Maurya et al. explores how MR tools can be integrated into the design process to enhance user participation and creativity, particularly in the early stages of product development [14]. The study highlights the advantages of MR environments in enhancing user interaction with 3D models and shows how MR tools can overcome the limitations of traditional 2D displays by providing an

immersive environment where depth perception is more natural and intuitive.

All in all, most research indicates a clear advantage of the usage of stereoscopic displays for scenarios where the user would otherwise be dependent on a two-dimensional form of presentation. What has not been investigated in any of the previously mentioned research, is the immediate effect that enhanced depth perception could have on users on an expert level in the field of 3D graphics design. Furthermore, an extension of a 2D display with digital 3D objects has not been discussed. This raises the question, how experts level users in the field of 3D graphics design would react to a combination of 2D screen and mixed reality enhancement.

3 Research Question and Hypothesis

As outlined above, there remains a critical gap in understanding how expert 3D modelers (those who routinely work with complex, detailed models) can benefit from integrating MR tools into their traditional 2D screen-based workflows. Specifically, how might MR affect depth perception and spatial understanding in ways that compensate for the limitations of 2D displays? Additionally, the potential for combining 2D and 3D views in a single workspace has not been fully explored. This research aims to fill these gaps by examining how expert 3D modelers respond to a dual-environment setup, integrating MR into their established workflows and evaluating the impact on user satisfaction, task performance, and spatial awareness. The primary research question driving this study is:

RQ1: How does the integration of Mixed Reality tools into traditional 3D modeling software affect the user's depth perception, task efficiency, and satisfaction compared to a 2D screen setup?

From this central question, several sub-questions emerge:

- **RQ1.1:** Does using MR tools reduce the time required to complete complex 3D modeling tasks?
- **RQ1.2:** How does MR influence the accuracy of spatial and depth-related judgments during the modeling process?
- **RQ1.3:** Do users report higher levels of satisfaction when using MR for 3D modeling compared to traditional 2D interfaces?
- **RQ 1.4:** What should an application and its user interface offer to let the user benefit the most from combining MR with their traditional workflow?

To explore the above research questions, the following hypotheses are formulated:

- **H1:** Using an MR tool in 3D modeling will take the user the same amount of time, compared to relying solely on traditional 2D screen setups.
- **H2:** Users will exhibit higher satisfaction levels with MR tools than with traditional 2D setups, as measured by a user satisfaction scale.
- **H3:** MR tools will lead to fewer errors in spatial judgment tasks, such as object placement and alignment, compared to 2D screen-based modeling.

4 Prototype Development

4.1 Objective of the Prototype

The primary goal of the prototype was to integrate MR into the traditional 3D modeling workflow, allowing users to seamlessly observe and interact with the 3D models they were working on in a Mixed Reality environment while still maintaining the familiar structure of a 2D monitor setup. By enabling this dual-environment setup, the prototype sought to address a major challenge in 3D modeling—limited depth perception on 2D screens—while prioritizing user comfort and minimizing disruption.

The prototype was designed with the hypothesis that providing an immersive MR view alongside the standard 2D monitor workflow would address core issues in traditional 3D modeling, such as spatial misjudgments, and ultimately improve user satisfaction. By allowing modelers to see and work on their 3D designs with greater perceptual accuracy, the prototype aimed to enhance user engagement, task efficiency, and the overall user experience.



Figure 2: An overview of the 3D modeling setup from user perspective showing a 3D model flying in the air, the same model on a virtual monitor in blender and the real environment of the user.

4.2 Hardware and Software Setup

The prototype system was developed using a combination of hardware and software tools to create an integrated MR environment compatible with existing 3D modeling workflows.

Hardware Configuration:

- **Windows Desktop:** The main workstation was a high-performance Windows desktop, equipped to handle virtual reality requirements efficiently. Exact specifications were flexible as long as the system could reliably run MR applications.
- **Meta Quest 3 Headset:** This Mixed Reality headset was chosen for its accessible passthrough capabilities, which allowed users to visualize their 3D models in the physical workspace.
- **Stable USB 3 Cable Connection:** A USB 3 cable ensured a stable, low-latency connection between the Meta Quest 3 headset and the desktop computer.

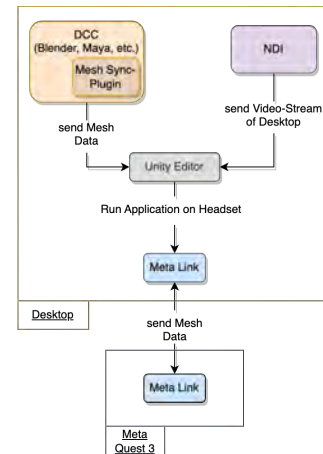


Figure 3: System architecture of the main components used for the prototype. A desktop PC is running a DCC (in this case blender) with Mesh Sync-Plugin, NDI, Unity and Meta Link, that sends mesh data to a Quest 3 device.

- **Standard Peripheral Setup:** Traditional peripherals, including a keyboard, mouse, and monitor, were included to allow users to maintain a familiar 3D modeling workflow while accessing MR functionalities.

Software Configuration:

- **Unity:** Unity served as the central platform for rapid prototyping and connecting the Meta Quest 3 headset. This environment allowed the integration of various tools and plugins needed for Mixed Reality functionality.
- **Mesh Sync by Unity:** Unity's MeshSync is a synchronization package that facilitates real-time updates of 3D models between DCC (Digital Content Creation). It supports Blender, Maya, 3ds Max and Motion Builder. By connecting Unity to these DCC tools via a server (created using the MeshSync-Server component), users can instantly visualize changes made to models in Unity as they are edited in real-time. It synchronizes mesh data into the MR environment, enabling seamless model updates in the MR headset. This plugin was critical for preserving the integrity of users' primary modeling workflows.
- **NDI (Network Device Interface):** NDI is a video connectivity technology that enables low-latency, high-quality video and audio transfer over standard IP networks. It allows devices within the same network to share video in real-time without needing specialized hardware. Designed for flexible use across industries, NDI supports resolutions up to 4K and includes advanced features like bi-directional transmission, interoperability across devices, and plug-and-play simplicity. It's commonly used in broadcasting, conferencing, gaming, and professional AV environments. In our prototype it was used to capture and stream the computer screen into Unity, enabling real-time viewing within the Meta Quest 3. This

setup allowed users to view the modeling workspace on a virtual monitor displayed alongside their 3D models, creating a cohesive 2D + MR experience.

- **Blender:** Blender was used as the primary 3D modeling software, with support extended to other tools like Maya and 3ds Max through Mesh Sync. The Mesh Sync plugin provided flexibility in software choice, allowing users to work with their preferred 3D modeling software while accessing MR capabilities.

4.3 Core Features and Functionalities

The prototype was built to incorporate key features that enhance the 3D modeling workflow by seamlessly integrating MR functionalities into traditional DCC environments. These core functionalities include:

Uploading 3D Models from DCC Tools The prototype supports uploading 3D models from commonly used DCC tools (Blender, Maya, and 3ds Max) directly into the MR environment. This seamless integration bridges the traditional 2D screen and MR experience, enabling users to view, manipulate, and edit models intuitively within a spatial context.

Real-Time Editing and Display of Changes Users receive real-time visual feedback on any modifications made within Blender, which is streamed directly to the MR environment through Unity's Mesh Sync plugin. This critical feature allows for instant updates to models, though the performance demand increases with model complexity. To manage these resource demands, a "Safe Performance" toggle was implemented, allowing users to control the update rate and reduce performance lag.

Interactive Model Manipulation The MR environment enables users to interact with 3D models directly using hand gestures, providing natural, intuitive control over model transformations, including translation, rotation, and scaling. For example, models can be scaled by adjusting hand distance, offering a responsive, tactile sense of control that closely mirrors real-world manipulation.

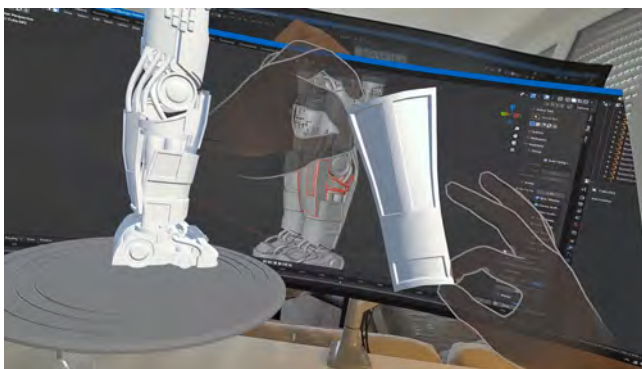


Figure 4: The user is scaling a 3D model in 3D space by using both tracked hands and a gesture.

Animation Playback and Editing in 3D The prototype facilitates the real-time playback and editing of animations within the MR environment, allowing users to visualize animated models with true-to-life movement and depth. This feature is essential for those

working with dynamic models and contributes to an immersive, realistic modeling experience.



Figure 5: The prototype uses an interactable turntable on which the 3D model is presented to the user. The turntable can be grabbed by the user to rotate the model in space.

Turntable The turntable serves as a central reference point in the MR space, grounding the model and attached UI elements for consistency and easy access. This feature aids in reassembling model parts and maintaining model orientation, which is crucial for spatial organization in complex projects.

UI Menu with Core Controls The UI menu, attached to the turntable, provides quick access to essential controls, including:

- (1) **Scale Adjustment:** Users can scale the model from 0.01x to 1.0x of the original size.
- (2) **Turntable Rotation Toggle:** This function allows users to start or stop the turntable rotation for easier viewing of different model angles.
- (3) **Safe Performance Mode Toggle:** This toggle allows users to activate performance-saving settings when working with complex models in Blender.
- (4) **Virtual Monitor Toggle:** Users can activate or deactivate a virtual monitor that mirrors the computer's display in the MR environment, providing convenient access to traditional tools alongside the MR workspace.

The UI menu can be minimized with a button on the turntable, allowing for an unobstructed workspace while retaining easy access to essential controls.

Virtual Monitor This feature projects the user's desktop view within the MR environment, giving users access to the computer's display without breaking immersion. It enables users to work with traditional 3D modeling software, maintaining their established workflows while benefiting from MR's spatial advantages.

Hand-Based Interaction All interactions within the MR environment are performed using hand gestures, allowing users to transition smoothly between the MR environment and traditional 2D tools like the computer mouse. Although direct MR-to-mouse interaction was not implemented, the focus on hand-based control minimizes the need for separate MR controllers and maintains user familiarity with their regular workstation setup.

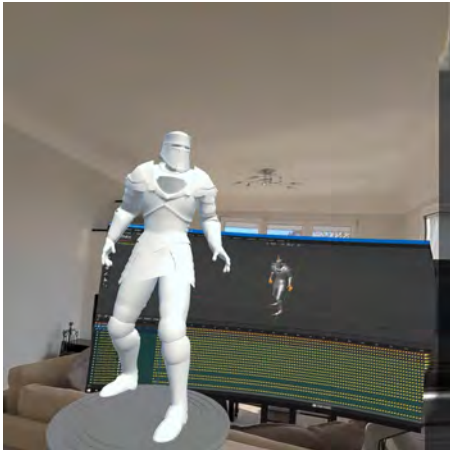


Figure 6: User viewing a 3D animation displayed on the virtual monitor within the mixed reality environment.

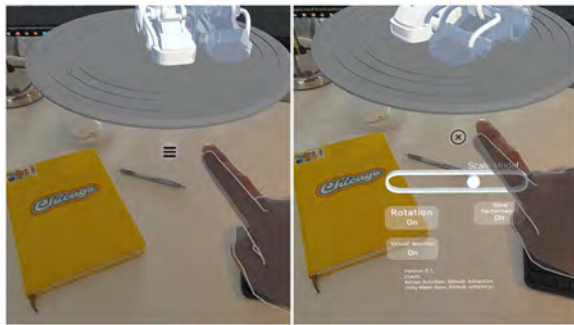


Figure 7: The 2D user interface with advanced options and settings can be switched from hidden (left) to extended (right).

4.3.1 Summary of Core Features and Functionalities. In sum, the prototype provides a hybrid workflow by integrating 3D modeling capabilities with MR features that include uploading and real-time model manipulation, interactive animation playback, an adjustable turntable, and an intuitive UI for core controls. The use of hand gestures for MR interaction aligns well with traditional computer setups, supporting a natural transition between 2D and MR environments. This setup provides enhanced spatial awareness and ease of manipulation, making MR a valuable tool in complex 3D modeling workflows.

4.4 Workflow Integration and User Interface

The prototype is designed to complement, not replace, traditional 3D modeling workflows, addressing specific limitations related to depth perception and spatial understanding without disrupting established practices. It allows modelers to retain their standard 2D workflow with a monitor, mouse, and keyboard setup, while providing an immersive MR environment to enhance real-time depth perception.

Seamless Transition Between 2D and MR: Users can move between 2D and MR views without losing focus on their primary

modeling tasks. By displaying 3D models in real time within MR, the prototype gives modelers an intuitive sense of scale and depth, which is difficult to achieve using only 2D screens. This feature provides a clear benefit for tasks that require precise spatial judgments, allowing users to identify potential issues in object proportions or positioning as they work.

Customizable UI Positioning: The MR interface offers flexibility for user comfort and focus by allowing the UI menu and controls to be moved to any convenient location within the MR space. This helps modelers avoid unnecessary interruptions, as the menu can be positioned within easy reach without obstructing their view of the model or the computer screen.

Efficient Control Mechanisms: To streamline the user experience, specific shortcut mechanisms have been implemented. For instance, rather than requiring users to open the menu to start or stop the turntable rotation, a quick double-tap on the turntable itself toggles the rotation. This feature not only saves time but also helps users remain immersed in their modeling tasks, as they do not need to navigate through additional UI layers for common actions.

Compatibility with Traditional Workflows: The prototype respects the modeler's need for keyboard shortcuts and mouse navigation within the DCC software. By enabling quick access to MR features without breaking from their 2D setup, it supports the modeler's ability to work comfortably with both traditional and MR tools side by side. Additionally, no major changes to the typical 3D modeling environment were made, so users can use the prototype as an extension of their current workflow rather than an entirely new system.

4.5 Challenges and Solutions

The development of the prototype faced several key challenges, particularly in optimizing system performance and overcoming compatibility issues with third-party tools like Mesh Sync. Below is a summary of the primary obstacles encountered and the solutions implemented to address them.

1) Performance Optimization

A major challenge was ensuring the system maintained high performance in a mixed reality environment while keeping models interactive (scalable, grabbable, etc.). High computational demands on the system, especially with complex models, led to potential motion sickness and hindered user experience. Key taxing processes are the following:

Real-Time Data Transfer The process of sending model data from the DCC (e.g., Blender) to the MR environment was resource-intensive. Each model had to be exported, imported into Unity, and made interactive by attaching and adjusting components automatically.

Real-Time Editing To allow real-time display of any changes made to the 3D model, the system continuously updated model data. This constant updating heavily taxed system resources, especially when changes occurred several times per second.

To mitigate these performance issues, two solutions were implemented: **(1) Selective Model Updates:** The system was optimized to update only the modified parts of the model rather than re-importing the entire mesh. This optimization reduced the frequency of large data transfers, significantly lowering processing demands.

(2) Safe Performance Mode: A “safe performance” mode was introduced to address performance drops associated with Unity’s mesh colliders. While mesh colliders allow precise grabbing and interaction, they require substantial computing power, recalculating multiple times per second. Safe performance mode replaced mesh colliders with box colliders, which, while less precise, dramatically reduced the computational load, stabilizing the frame rate from a drop to 7 fps back up to the recommended 70 fps on the Quest 3.

2) Adapting Mesh Sync for Custom Use

Another significant challenge was adapting Mesh Sync, a networking solution originally developed to synchronize mesh data across platforms. Using Mesh Sync for real-time updates in an MR environment was beyond its intended use. To integrate it effectively, complex plugin configurations and adjustments were required, including custom scripts to streamline synchronization between the DCC and MR environment.

3) Installation Constraints and Platform Limitations

Direct installation on the Meta Quest 3 headset was not feasible due to compatibility issues. The Mesh Sync plugin is designed for desktop platforms and is not optimized for the Quest 3’s Android-based operating system, which lacks support for some necessary libraries. This constraint necessitated several compromises: (1) *Operating Through Unity Play Mode:* The system operated in Unity’s Play Mode on a tethered desktop, rather than as a standalone application on the headset. This setup allowed the Unity editor to handle hand-tracking and rendering tasks but limited the MR experience by relying on the desktop for processing. (2) *Impact on Hand Tracking:* Running the system on the desktop rather than on the Quest 3 platform affected hand-tracking responsiveness, as hand-tracking calculations were processed on the desktop rather than natively on the headset. Despite this, operating in Play Mode had the advantage of reducing synchronization lag, as it avoided wireless transmission delays that would have resulted from mesh data transfers over a wireless connection.

5 Study Design

5.1 Expert Interviews

The goal of the user study is to assess the impact of integrating MR into traditional 3D modeling workflows. To gain a comprehensive understanding of this integration, semi-structured expert interviews were chosen as the primary data collection method. This approach is suitable for exploratory research, allowing for both guided questions and the flexibility to explore participants’ nuanced perspectives. Through these interviews, the aim is to uncover key insights into the technical trade-offs, workflow changes, and the potential benefits or limitations of incorporating MR into established 3D modeling practices. There was no ethical approval required by our institution.

5.1.1 Reasoning for Expert Interviews. Expert interviews were selected over broader user testing for several reasons:

- **Nuanced Perspectives:** Experts can provide detailed insights into how MR influences spatial understanding, workflow efficiency, and user satisfaction in professional settings.

They can articulate the specific challenges they face in traditional 3D modeling and how the new MR system addresses or exacerbates these challenges.

- **Familiarity with 3D Modeling Tools:** Experts in this field possess in-depth knowledge of both traditional 3D modeling software and newer MR tools. Their comparative insights between the two workflows will offer a more informed evaluation than general users could provide.
- **In-depth Exploratory Data:** The qualitative data from expert interviews will provide a richer, more exploratory perspective than quantitative methods alone. This is essential for identifying key themes like usability, technical challenges, and potential improvements in the MR system.

5.1.2 Expert Selection. Given the exploratory nature of the study, a sample of six experts was selected. This sample size allows for a manageable yet diverse set of insights enabling cross-comparison. The experts were recruited locally and included three professionals from 3D studios and three students of a master program on game design. The following criteria were used for expert selection:

- Participants must have at least 3 years of experience in 3D modeling. This ensures that they are well-versed in traditional workflows and can critically evaluate the addition of MR tools.
- Familiarity with AR/MR Technologies: While deep expertise in MR is not mandatory, participants should have some exposure to AR/MR systems, ensuring they can contextualize their feedback within a broader technological landscape.

5.2 Interview Procedure

The interview process includes the following steps:

- (1) **Introduction:** The participant is first introduced to the study’s goals and the MR system. This is followed by an initial familiarization phase, where the participant can explore the system without a specific task.
- (2) **Task Completion:** The participant is asked to complete two modeling tasks—one using only a 2D monitor (referenced as “2D”) and one using the MR-enhanced system with the 2D monitor (referenced as 2D + MR).

For each participant, the order of which task (2D vs. 2D + MR) is being altered to ensure the results are not biased by the order in which tasks are administered. In this study, the first user starts with the 2D + MR system. This means three participants started with the MR-enhanced system, while the other three began with the traditional 2D workflow.

The tasks involve modeling an American-style mailbox, a simple object chosen for its familiarity and ease of interaction. The time taken to complete each task is recorded.

- (3) **EUCS Questionnaires:** After completing each task, the participant is given the End User Computing Satisfaction (EUCS) questionnaire. This survey measures user satisfaction across five dimensions: content, accuracy, format, ease of use, and timeliness. A Likert scale is used to capture their ratings.
- (4) **Post-Task Questionnaire:** After completing both tasks and questionnaires, the participant has the opportunity to share more about their opinions and experiences, if they haven’t

already done so during the introduction or task. Participants were asked the following post-task questions:

- **Challenges in Current Workflows:** What challenges do you currently face in your 3D modeling workflow, and how did the MR system address these challenges?
- **Potential Benefits:** How did the MR system improve your workflow in terms of spatial understanding, task efficiency, or design accuracy?
- **Barriers to Adoption:** What limitations or frustrations did you experience with the MR system? What would prevent you from adopting this tool in your professional workflow?
- **Suggestions for Improvement:** How could the MR system be improved to better fit your needs? What additional features or adjustments would make it more useful in your work?

During the interviews observations, quotes and expressions were noted in form of qualitative insights using thematic analysis, focusing on recurring themes such as workflow efficiency, usability, and perceptual improvements.

5.3 Task Design

The tasks assigned to the participants were carefully designed to evaluate the impact of MR on 3D modeling workflows. The task involves creating a foreign but familiar object—a standard American-style mailbox—using reference images, as seen in Fig. 8. This object was selected for its simple topology and universal recognizability, which ensures that all participants, regardless of their specific 3D modeling background or any other factors, are familiar with its form and can focus on modeling accuracy and spatial perception rather than conceptualizing the object.



Figure 8: These reference images were given to the participants for the 3D modeling task.

6 Results and Discussion

6.1 Task Completion Times

6.1.1 Presentation of Results. The task completion times for both the 2D and 2D + MR setups are summarized below. These results

give us a direct comparison of how long participants took to complete the 3D modeling tasks under each condition.

Condition	Min (min.)	Max (min.)	Mean (min.)	Std. Dev. (min.)
2D	6:13	20:24	13:36	5:31
2D + MR	13:37	50:00	21:51	12:21

Table 1: Summary of Time Measurements for Different Conditions

It is important to note that the maximum time of 50 minutes for the 2D + MR condition was due to one user testing the system and extensively discussing the experience with the tester during the task. This outlier likely skewed the average for the 2D + MR condition.

6.1.2 Interpretation. From the results, it is clear that participants were not faster using the 2D + MR setup compared to the 2D setup. The mean task completion time for 2D + MR was notably longer (21:51 minutes) compared to 2D (13:36 minutes). Several factors could explain this difference:

- **Learning Curve:** Participants needed time to familiarize themselves with the MR system. During the qualitative interviews, it was observed that users had difficulty adjusting to the unfamiliar computer setup. This included navigating the new hand-tracking system, testing the capabilities of MR, and interacting with the models in a different environment.
- **Experimentation:** Some participants spent additional time exploring the features and capabilities of the MR system during the task, which further extended task completion times. For example, one participant took significantly longer as they were testing out the system while performing the task.

6.1.3 Link to Hypotheses. These findings directly address H1, which states that “Using an MR tool in 3D modeling will take the same amount of time, compared to relying solely on traditional 2D screen setups.”

Based on the results, H1 can be rejected. The task completion times were not the same for the 2D and 2D + MR setups, with the 2D + MR setup taking longer on average. The longer completion time is likely due to the learning curve and participants’ additional experimentation with the MR system.

Further refinement of the MR tool and providing users more time to familiarize themselves with the system could potentially reduce task completion times in future studies.

6.2 User Satisfaction (EUCS Scores)

6.2.1 Presentation of Results. The results from the EUCS questionnaire, summarized in Table 2, indicate the mean scores for each dimension across both setups.

In the Content category, both setups showed high satisfaction scores, with the 2D setup comparable to the 2D + MR setup (Mean = 4.28 for 2D vs. 4.06 for 2D + MR). Similarly, for Accuracy, scores were close, with the 2D setup at a mean of 4.06 and the 2D + MR setup at 4.00, both showing moderate variability in user responses (Std. Dev. = 0.938 and 0.970).

Dimension	Setup	Mean	Std. Deviation
2*Content	2D	4.28	0.669
	2D + MR	4.06	0.802
2*Accuracy	2D	4.06	0.938
	2D + MR	4.00	0.970
2*Format	2D	4.39	0.778
	2D + MR	4.11	0.963
2*Ease of Use	2D	3.94	1.056
	2D + MR	3.72	1.018
2*Timeliness	2D	4.22	1.060
	2D + MR	3.28	1.127
Total	2D	4.18	0.907
	2D + MR	3.83	1.008

Table 2: Mean EUCS scores for each dimension and setup (2D vs. 2D + MR)

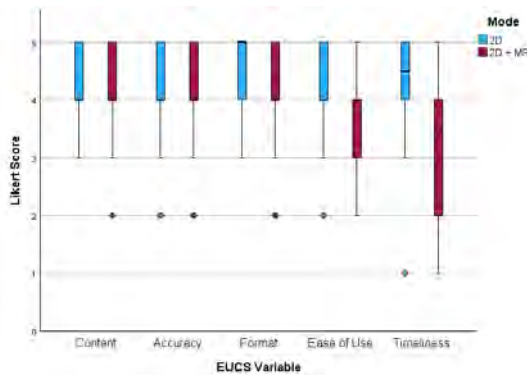


Figure 9: Comparison of Mean EUCS Scores between 2D and 2D + MR across the Five EUCS Dimensions

In Format, the traditional 2D setup achieved a mean score of 4.39, indicating strong user satisfaction. The 2D + MR setup also scored well at a mean of 4.11, with a distribution of responses (Std. Dev. = 0.963).

The Ease of Use and Timeliness dimensions displayed more noticeable differences. The 2D setup showed a higher Ease of Use score with a mean of 3.94 compared to 3.72 for the 2D + MR setup, suggesting users found the traditional setup slightly easier to navigate. Timeliness reflected the largest contrast, with a mean score of 4.22 for the 2D setup and a lower mean of 3.28 for the 2D + MR setup, indicating a marked difference in perceived efficiency between the setups. The standard deviations for Timeliness (1.060 for 2D, 1.127 for 2D + MR) reveal this variation is consistent across participants.

Overall, total satisfaction scores show a mean of 4.18 for the 2D setup and 3.83, as seen in Table 2, for the 2D + MR setup, indicating a slight preference toward the traditional 2D workflow.

6.2.2 Interpretation. The Content and Accuracy dimensions showed minor differences between the 2D and 2D + MR setups, with both setups receiving high ratings, indicating that users found both systems adequate in these aspects. The Content score, in particular,

was slightly higher for the 2D setup, possibly due to familiarity and efficiency when using well-established workflows on a traditional screen.

The Format dimension showed a slightly higher score for the 2D setup (mean = 4.39) compared to the 2D + MR setup (mean = 4.11), suggesting that users found the interface and presentation of information somewhat easier to interpret in the traditional setup. This may relate to the relative novelty of the 2D + MR system, the virtual monitor setup inside the 2D + MR system and the potential need for additional refinement in interface customization.

Ease of Use and Timeliness revealed the most contrasts, with users rating the 2D setup higher in both aspects. Ease of Use scores may reflect initial difficulties in adapting to the hand-tracking and input mechanisms in the MR environment, as noted in user feedback regarding false positives and challenging keyboard interactions. Timeliness scored notably lower for the 2D + MR setup, which aligns with participants' reports of setup complexities and additional time required to perform tasks.

A noteworthy outlier is visible in Fig. 9 under the Ease of Use dimension, where one participant rated the questions "The feedback provided by the 2D + MR system during my modeling tasks was timely" and "The feedback provided by the 2D system during my modeling tasks was timely" at a 1. This could be due to a misunderstanding of the questions, as they allow some room for misinterpretation. No other participant chose a score this low.

6.2.3 Link to Hypotheses. Hypothesis H2 posits that users will be more satisfied with the MR-enhanced system compared to the 2D setup. The results, however, do not support H2, as the 2D setup scored consistently higher in each EUCS dimension. The observed differences in Ease of Use and Timeliness suggest that while MR may offer valuable depth perception benefits, as reflected in qualitative feedback, it may also introduce friction in usability that reduces overall satisfaction in a professional workflow.

6.3 Workflow Efficiency and Task Performance

One of the most notable findings from the expert interviews was the general consensus that the 2D + MR system enhanced spatial understanding, which led to more efficient task performance in some cases. One participant said that "I am paying more attention to details" and another one "I just need to turn my head and it (the model) is there". A third one supported this by saying "Viewing from an arbitrary perspective is pretty awesome". Several participants commented that they were able to reduce the need for constant orbiting around their models, a common strategy when working solely with 2D screens, indicating a reduction in the cognitive load associated with constantly reorienting the model. This suggests that the 2D + MR setup, by allowing users to view their models in a more intuitive 3D space, helped improve efficiency by minimizing unnecessary actions. One participant commented on that, "it's better than having 2 monitors".

However, some participants found that the 2D + MR system did not always offer smooth task performance due to technical issues, particularly with the passthrough feature. For instance, one participant suggested that Meta's prototype "Orion" would have been a better choice of headset, instead of the Meta Quest 3, to increase comfort and avoid reliance on passthrough, which hindered their

workflow in certain tasks. By the time of publishing this work, the prototype mentioned had not been publicly available for testing.

6.3.1 Spatial Understanding and Depth Perception. A central hypothesis of this study, H3, was that the 2D + MR system would enhance spatial understanding and depth perception during 3D modeling tasks. The interviews confirmed this, with participants consistently noting improvements in their ability to gauge size relations and spatial accuracy.

- **Improved Perception:** One participant explained, “*Seeing it in reality helps a lot to understand size relations,*” highlighting how 2D + MR helped bridge the gap between the flat 2D screen and their real-world expectations. The ability to assess their models in a more immersive way allowed them to identify errors in size and proportion that were not as easily detected in the 2D environment.
- **Error Detection:** Another participant noted that the 2D + MR system allowed them to detect issues in their models that they would not have noticed otherwise, suggesting that MR improves error detection, particularly in spatial tasks.

These findings align with hypothesis H3, which states, “*MR tools will lead to fewer errors in spatial judgment tasks, such as object placement and alignment, compared to 2D screen-based modeling.*”

6.4 Usability and User Experience

While participants appreciated the benefits of 2D + MR in terms of spatial awareness, there were clear usability challenges that affected the overall user experience. Hand tracking and keyboard use in the passthrough environment were cited as major pain points.

- **Hand Tracking Issues:** Multiple participants reported problems with the accuracy of the hand tracking system, which led to false positives and hampered task performance. One participant stated that “*the hand tracking produced lots of false positives,*” causing frustration during their workflow. Although some of these challenges were expected, their negative impact on user satisfaction cannot be ignored.
- **Keyboard Interaction:** Participants also struggled with using a keyboard in the 2D + MR environment, noting difficulties in managing both virtual and physical elements simultaneously. A recurring issue was that participants had to use a provided setup, preventing them from accessing their personal computer environments, including preferred keyboard layouts, custom plugins, and individual software settings. This issue is critical because traditional modeling workflows rely heavily on keyboard shortcuts and precise input, which were difficult to replicate in the MR setup.

Additionally, comfort was a concern for many participants. Several found the MR headset uncomfortable after prolonged use, which could affect the long-term viability of MR tools in professional 3D modeling environments. This highlights the need for ergonomic improvements in MR devices to support extended work sessions.

6.5 Barriers to Adoption

Several barriers to adopting the 2D + MR setup in experts' workflows were identified:

- **Passthrough Quality:** The resolution of the virtual monitor and the quality of the passthrough video were commonly cited as issues that reduced the effectiveness of the 2D + MR environment. These factors led to a more abstract, less intuitive experience when switching between the virtual and physical environments.
- **Setup Overhead:** Participants noted that the initial setup of the 2D + MR system was cumbersome, which would pose a significant barrier to its adoption in day-to-day professional tasks. The time required to adjust the 2D + MR environment detracted from its overall utility.
- **User Experience (UX) Improvements:** Users pointed out specific UX improvements that could enhance the system's usability, such as locking model axes, improving material and shader support, and adjusting the virtual monitor's size.

6.6 Potential Benefits Identified by Participants

Despite the usability challenges, participants recognized clear benefits that 2D + MR could bring to 3D modeling workflows:

- **Enhanced Error Detection:** The ability to quickly identify and correct errors in real-time was a significant advantage noted by several participants. This suggests that 2D + MR systems could reduce post-processing time in professional workflows by enabling users to identify and resolve issues during the modeling phase, which would strongly support hypothesis H3: “*MR tools will lead to fewer errors in spatial judgment tasks, such as object placement and alignment, compared to 2D screen-based modeling.*”
- **Better Depth Perception:** Users praised the immediate feedback provided by 2D + MR, which allowed them to visualize models in a real-world context. One participant noted that MR made it easier to visualize their work in Unity, suggesting that MR could streamline workflows for creating content for VR applications. Both the enhanced error detection and improved depth perception observed in the study could further validate H3.

Another notable observation came from a participant who highlighted that the ideal use case for the 2D + MR system is in sculpting tasks, such as creating faces, monsters, or player characters—projects where reference blueprints are less effective. In these scenarios, precise spatial understanding is critical.

Additionally, multiple participants noted that the 2D + MR system could address issues with incorrect focal length and resulting depth perception discrepancies, a common problem in Blender. This mismatch often occurs when the model looks different in Blender's viewport, due to virtual camera settings, compared to how it appears in reality or VR, where natural depth perception and accurate field of view provide a more realistic sense of scale.

6.6.1 Key Suggestions for Improvement. Participants provided several suggestions for improving the 2D + MR system:

- (1) **Hand Tracking Improvements:** Improving the precision and reliability of hand tracking was a common request, as many users found the current system too inaccurate for professional work.

- (2) **Customization and Flexibility:** Many participants suggested additional customization options, such as the ability to synchronize materials or textures to the models, use random MatCaps for better visual differentiation, and display the model in wireframe mode, enhancing alignment with their specific workflows.
- (3) **Simplified Workflow Integration:** Features like hand gesture-based model rotation and tap-to-delete functionality were mentioned as potential improvements that would make 2D + MR tools more fluid and easier to integrate into existing workflows.

Observations of participants' usage, along with their feedback, help pinpoint specific scenarios where experts would benefit most from a 2D + MR system:

- (1) **Modeling with Minimal Reference:** The ideal use case for the 2D + MR system is in sculpting tasks, such as creating faces, monsters, or player characters—projects where reference blueprints are less effective. In these cases, precise spatial understanding is critical, making 2D + MR tools particularly valuable.
- (2) **Quickly Showcasing 3D Models:** Participants emphasized the benefit of being able to quickly showcase 3D models to others, especially non-experts like clients. The 2D + MR setup provides an intuitive way to present designs and receive immediate feedback.
- (3) **Working on Complex Models with Intricate Shapes:** Some experts noted that the 2D + MR system would be especially useful for more complex 3D modeling projects they've encountered in the past. This was specifically highlighted in contrast to the simplicity of the 3D model used during the user study.

6.7 Conclusion

This prototype-driven research proposes the potential of 2D + MR tools to transform 3D modeling workflows, especially the qualitative data from the expert interviews is valuable regarding spatial understanding, depth perception, and error detection. However, significant challenges related to usability, particularly concerning input methods (e.g., hand tracking, keyboard usage), need to be addressed for 2D + MR to be seamlessly integrated into professional environments. The feedback gathered from the interviews indicates that while the 2D + MR system has the potential to enhance workflow efficiency and accuracy, more development is needed to optimize the user experience and reduce the setup complexity for everyday use. Addressing these issues will be key to making the combination of MR with 2D screens a more viable tool for professional 3D modelers.

Furthermore, testing the prototype with a larger and more diverse audience might gain more useful insights for the further development. Additionally, long-term effects need to be taken into consideration. Experts in 3D modeling are trained in the traditional 2D approach for years. Hence, longer usage of the 2D+MR prototype could affect their performance. And finally, the concrete task might also have an influence on which technique fits and performs better. Experts mentioned that sculpting could be particularly useful with 2D+MR.

6.8 Addressing Hypotheses and Research Questions

H1: Task Completion Time Hypothesis H1, which proposed that task times for the 2D + MR system would be similar to the 2D setup, is not supported. The qualitative feedback, indicating a need for users to familiarize themselves with the system and experiment with its capabilities, further supports rejecting this hypothesis.

H2: User Satisfaction H2, hypothesizing higher satisfaction levels with MR tools, is also not supported by the EUCS scores, as satisfaction ratings were consistently higher for the 2D setup. However, qualitative data highlights MR's benefits for spatial understanding and error detection, suggesting that while overall satisfaction may be lower, the MR system offers specific advantages in tasks requiring depth perception, which could lead to higher satisfaction with further refinement.

H3: Spatial Judgment H3, predicting fewer spatial errors with MR, is supported. Participants reported improved spatial understanding and error detection in MR compared to 2D alone. This aligns with qualitative feedback that MR provides enhanced depth cues, making spatial judgment tasks more accurate.

Research Questions The results indicate that while MR may not yet streamline overall task efficiency as much as 2D, it holds unique advantages in spatial accuracy and modeling precision. Specific benefits included enhanced depth perception and situational awareness, particularly valuable in complex tasks like sculpting or model alignment.

6.9 Limitations of the Study

Sample Size and Generalizability This study's findings are based on a small sample size of six expert 3D modelers, limiting the generalizability of the results. With a larger and more diverse participant pool, future studies could provide a broader understanding of MR's impact across different user demographics and industries.

Lack of Familiarity with Equipment Participants used provided systems rather than their personalized work environments. As a result, they lacked access to their preferred keyboard layouts, plugins, and custom shortcuts, which could have led to a less efficient and satisfying experience than if they had used their own setups. This limitation highlights the importance of allowing experts to use familiar tools, to capture a more accurate representation of the technology's impact.

Task Design and Complexity The study's task—creating a simple model, such as a mailbox—may not fully showcase the strengths and weaknesses of MR. Modeling tasks that require complex spatial reasoning and intricate detail would likely provide a better test of MR's advantages in depth perception and spatial awareness. Future studies should consider using tasks more reflective of real-world challenges to better evaluate the effectiveness of MR.

Learning Curve with the MR System Although all participants were experienced with MR technologies, adapting to the specific MR setup used in this study presented challenges. Differences in software configurations, system setup, and interaction methods may have initially impacted their workflow efficiency and satisfaction. Future studies could benefit from a longer acclimation period or a longitudinal approach to capture how extended use of the specific

MR system affects task performance, satisfaction, and workflow integration.

Focus on Short-Term Performance This study focused on immediate usability and task efficiency, assessing only short-term outcomes. To provide a more comprehensive picture, future research should investigate the potential long-term productivity gains, learning effects, and sustained user satisfaction associated with MR tools.

By recognizing these limitations, future research can build upon this study to address these challenges and further support the effective adoption of MR in professional 3D modeling environments.

7 Conclusion

This paper explored the integration of 2D screens and Mixed Reality to enhance 3D modeling workflows, focusing on improvements in depth perception, spatial understanding, and user satisfaction. Our findings indicate that the 2D + MR setup shows promise for 3D modeling professionals, notably by enhancing spatial awareness and minimizing errors in model alignment and positioning.

A mixed-methods approach, combining quantitative metrics like task completion times and EUCS scores with qualitative feedback from expert interviews, demonstrated MR's potential for specific tasks. Notably, the hypothesis concerning spatial accuracy (H3) was supported, while hypotheses around task completion time and overall satisfaction yielded mixed outcomes. These results underscore the importance of further refining MR technology despite existing usability challenges.

The paper revealed exploratory insights into 3D modeling practices, presenting an innovative MR prototype that maintains familiar workflows while introducing enhancements to depth perception. The developed prototype offers an accessible and practical method to elevate modeling precision, marking a significant step forward in the field. Triangulating diverse data sources also enriched our understanding of the prototype's impact.

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